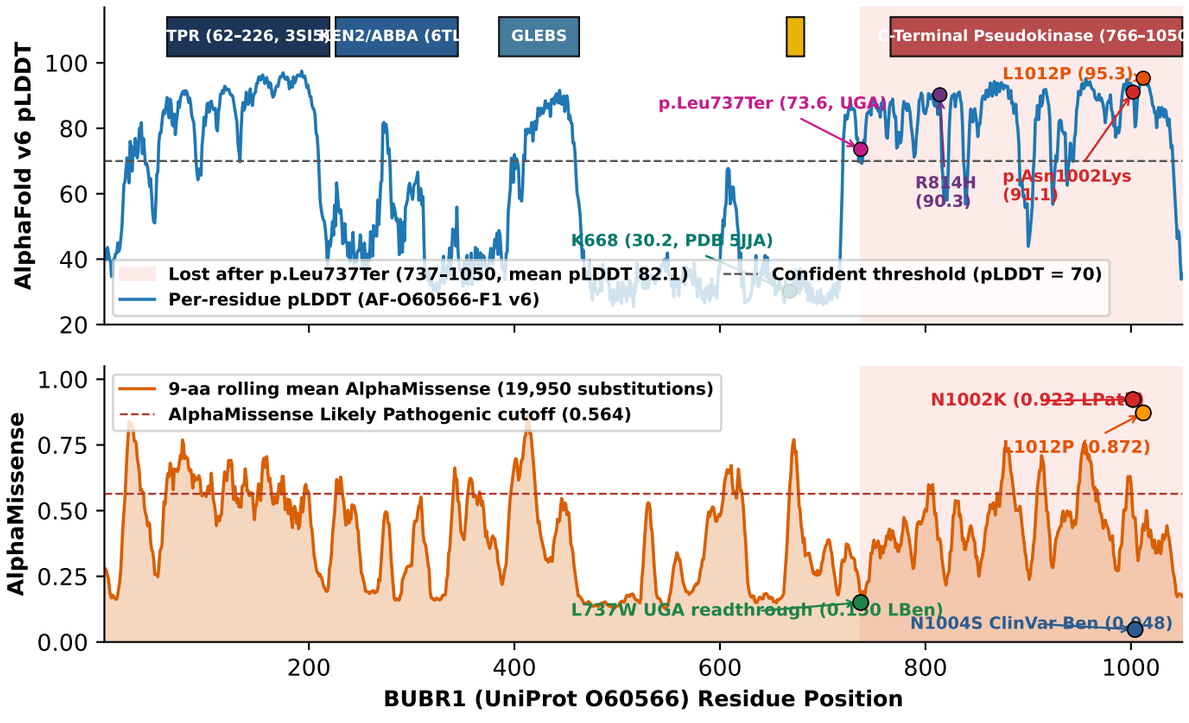
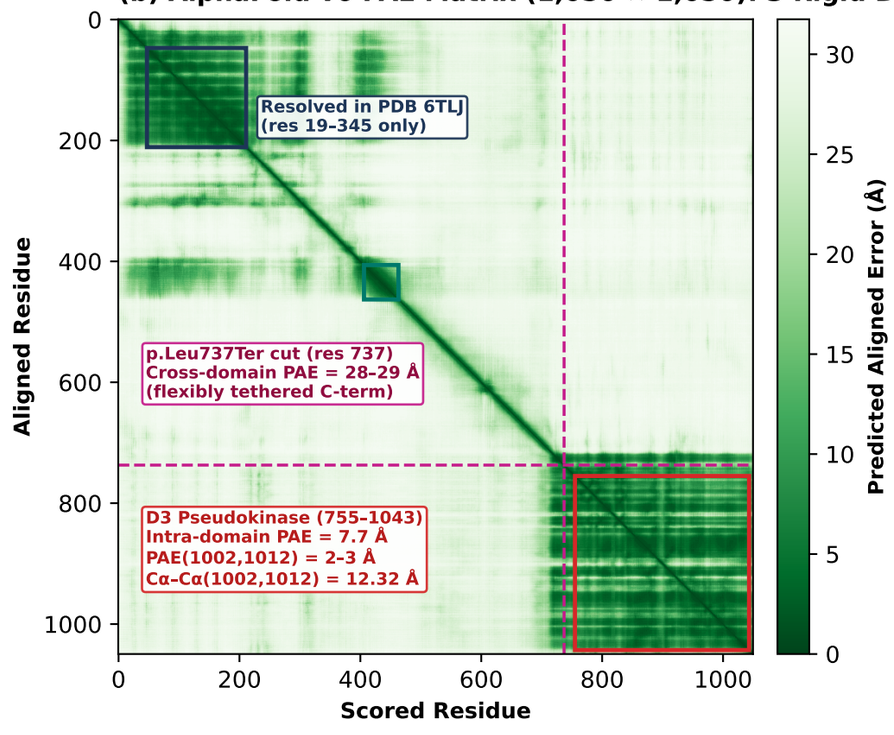
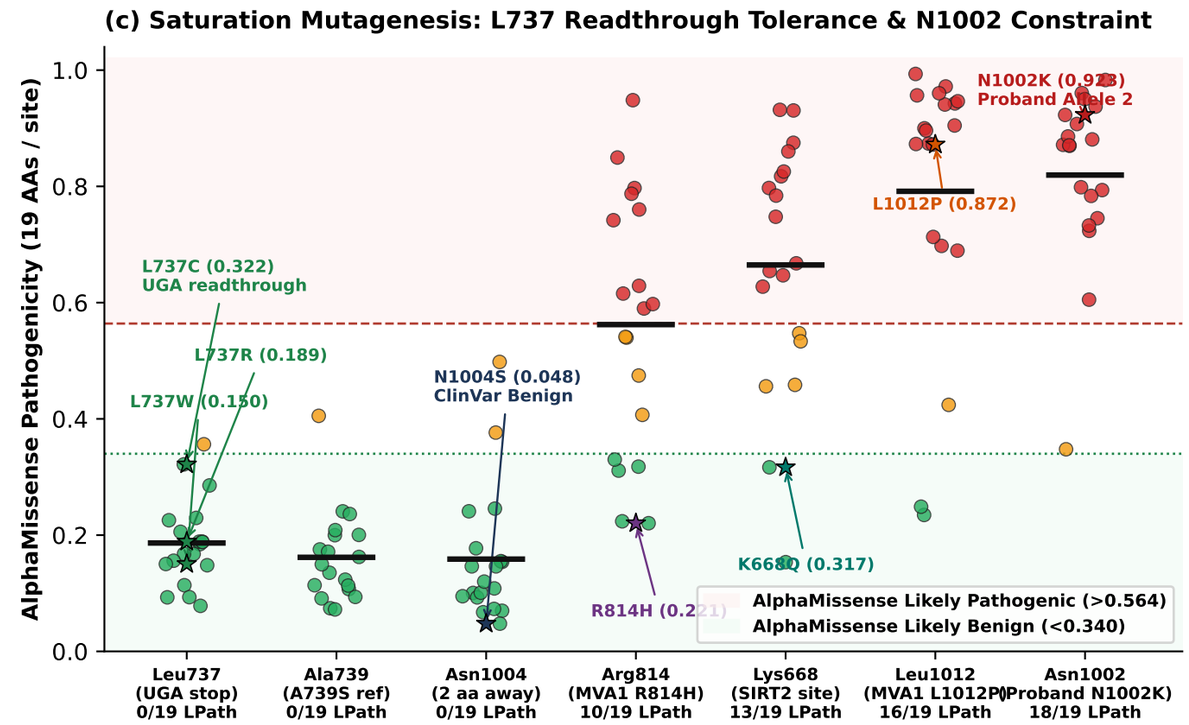
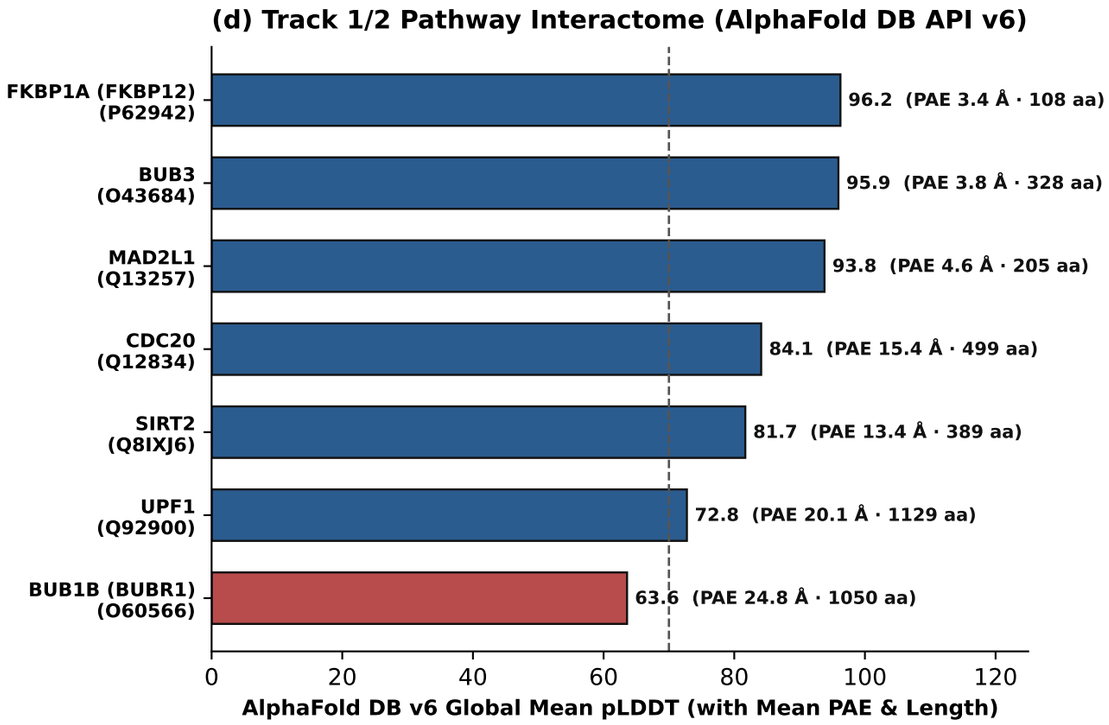



**ESM-1v 5-Model Ensemble Window Verification:**
- Window 29–1050 (canonical): N1002K = −0.110 vs L1012P = −1.801 (gap 1.69 nats; B/LB med −0.161)
- Window 1–1022 (fixed 2026-10-11): N1002K = −0.063 vs L1012P = −2.158 (gap 2.10 nats; B/LB med −0.177)

→ Both windows falsify H1 (N1002K ≠ L1012P in 1D sequence); AlphaMissense 0.9229 captures 3D C-cap H-bonds.